

Lab 5 Exercise — Task 1

Family Tree Knowledge Base



```
alhanoofamily.pl
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alhanoofamily.pl
male(ahmed).
male(ali).

female(sara).
female(nora).
% Gender Facts

male(ahmed).
male(ali).

female(sara).
female(nora).

% Parent Facts

parent(ahmed, ali).
parent(sara, ali).

parent(ahmed, nora).
parent(sara, nora).
% Rules

father(X, Y) :-
    male(X),
    parent(X, Y).
mother(X, Y) :-
    female(X),
    parent(X, Y).
son(X, Y) :-
    male(X),
    parent(Y, X).

daughter(X, Y) :-
    female(X),
    parent(Y, X).

brother(X, Y) :-
    male(X),
    parent(P, X),
    parent(P, Y).
```

```

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alhanoofamily.pl
sister(X, Y) :-
    female(X),
    parent(P, X),
    parent(P, Y),
    X \= Y.

grandfather(X, Y) :-
    male(X),
    parent(X, P),
    parent(P, Y).

aunt(X, Y) :-
    female(X),
    parent(P, Y),
    sister(X, P).

uncle(X, Y) :-
    male(X),
    parent(P, Y),
    brother(X, P).

cousin(X, Y) :-
    parent(P1, X),
    parent(P2, Y),
    parent(G, P1),
    parent(G, P2),
    P1 \= P2,
    X \= Y.

ancestor(X, Y) :-
    __parent(X, Y).

ancestor(X, Y) :-
    parent(X, Z),
    __ancestor(Z, Y).

```

```
54 ?- father(X, Y).
X = ahmed,
Y = ali ;
X = ahmed,
Y = nora ;
X = ahmed,
Y = ali ;
X = ahmed,
Y = nora ;
false.

54 ?- mother(X, Y).
X = sara,
Y = ali ;
X = sara,
Y = nora ;
X = sara,
Y = ali ;
X = sara,
Y = nora ;
false.

55 ?- mother(X, child_name).
false.

56 ?- son(X, Y).
X = ali,
Y = ahmed ;
X = ali,
Y = sara ;
X = ali,
Y = ahmed ;
X = ali,
Y = sara.

57 ?- son(X, parent_name).
false.
```

```
58 ?- daughter(X, Y).  
X = nora,  
Y = ahmed ;  
X = nora,  
Y = sara ;  
X = nora,  
Y = ahmed ;  
X = nora,  
Y = sara.  
  
59 ?- daughter(X, parent_name).  
false.  
  
60 ?- brother(X, Y).  
X = ali,  
Y = nora ;  
X = ali,  
Y = nora ;  
X = ali,  
Y = nora ;  
X = ali,  
Y = nora.  
  
61 ?- brother(X, person_name).  
false.  
  
62 ?- sister(X, Y).  
X = nora,  
Y = ali ;  
X = nora,  
Y = ali ;  
X = nora,  
Y = ali ;  
X = nora,  
Y = ali ;  
false.  
  
63 ?- sister(X, person_name).  
false.  
  
64 ?- grandfather(X, Y).  
false.
```

```
65 ?- grandfather(X, person_name).  
false.  
  
66 ?- aunt(X, Y).  
false.  
  
67 ?- aunt(X, person_name).  
false.  
  
68 ?- uncle(X, Y).  
false.  
  
69 ?- uncle(X, person_name).  
false.  
  
70 ?- cousin(X, Y).  
false.  
  
71 ?- cousin(X, person_name).  
false.  
  
72 ?- ancestor(X, Y).  
X = ahmed,  
Y = ali ;  
X = sara,  
Y = ali ;  
X = ahmed,  
Y = nora ;  
X = sara,  
Y = nora ;  
false.  
  
73 ?- ancestor(X, person_name).  
false.
```